

Exercise Series 2 (solutions)

Ex1 (a) We saw in the class that

$$\hat{\alpha} = \sum_{i=1}^n \left(\frac{1}{n} - w_i \bar{x} \right) y_i \quad \text{and} \quad \hat{\beta} = \sum_{i=1}^n w_i y_i,$$

$$\text{where } w_i = \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Since y_i 's are uncorrelated and $\text{Var}(y_i) = \sigma^2$, we set

$$\begin{aligned} \text{Var}(\hat{\alpha}) &= \sigma^2 \sum_{i=1}^n \left(\frac{1}{n} - w_i \bar{x} \right)^2 \\ &= \sigma^2 \sum_{i=1}^n \left(\frac{1}{n^2} + w_i^2 \bar{x}^2 - \frac{2}{n} w_i \bar{x} \right) \\ &= \sigma^2 \left\{ \frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right\} \left[\text{Recall } \sum_{i=1}^n w_i = 0 \right. \\ &\quad \left. \& \sum_{i=1}^n w_i^2 = \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \right] \\ \text{Var}(\hat{\beta}) &= \sigma^2 \sum_{i=1}^n w_i^2 = \frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}. \end{aligned}$$

(b) From the expression, minimizing $\text{Var}(\hat{\beta})$ is equivalent to maximizing $\sum_{i=1}^n (x_i - \bar{x})^2$, which is same as maximizing the spread of the data. This is achieved by placing the observations as far as possible, i.e., by choosing $\frac{n}{2}$ x_i values to be a and the remaining to be b .

A more rigorous justification for this is obtained from

$$\sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n} \sum_{i \neq j} (x_i - x_j)^2$$

Think about what needs to be done if you have to select one additional x value.

Ex 2 (a) We have seen that n

$$\hat{\alpha} = \bar{y} - \hat{\beta} \bar{x} \quad \& \quad \hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Now, $\hat{y}_i = \hat{\alpha} + \hat{\beta} x_i$

$$= \bar{y} + \hat{\beta} (x_i - \bar{x}) \Rightarrow \sum_{i=1}^n \hat{y}_i = n\bar{y} + \hat{\beta} \sum_{i=1}^n (x_i - \bar{x})$$

$$= \sum_{i=1}^n y_i$$

$$(b) \sum_{i=1}^n y_i^2 = \sum_{i=1}^n (\hat{y}_i + y_i - \hat{y}_i)^2$$

$$= \sum_{i=1}^n \hat{y}_i^2 + \sum_{i=1}^n (y_i - \hat{y}_i)^2 + 2 \sum_{i=1}^n \hat{y}_i (y_i - \hat{y}_i)$$

$$y_i - \hat{y}_i = y_i - \bar{y} - \hat{\beta} (x_i - \bar{x}), \quad \hat{y}_i = \bar{y} + \hat{\beta} (x_i - \bar{x})$$

$$\therefore \hat{y}_i (y_i - \hat{y}_i) = \{ \bar{y} + \hat{\beta} (x_i - \bar{x}) \} \{ (y_i - \bar{y}) - \hat{\beta} (x_i - \bar{x}) \}$$

$$= \bar{y} (y_i - \bar{y}) - \bar{y} \hat{\beta} (x_i - \bar{x}) + \hat{\beta} (x_i - \bar{x}) (y_i - \bar{y}) - \hat{\beta}^2 (x_i - \bar{x})^2$$

Observe that $\hat{\beta}^2 \sum_{i=1}^n (x_i - \bar{x})^2 = \hat{\beta} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$

This implies $\sum_{i=1}^n \hat{y}_i (y_i - \hat{y}_i) = 0.$

$$(c) \sum_{i=1}^n (y_i - \bar{y})^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2$$

$$\text{Similarly, } \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^n (\hat{y}_i - \bar{\hat{y}})^2 = \sum_{i=1}^n \hat{y}_i^2 - n\bar{\hat{y}}^2 = \sum_{i=1}^n \hat{y}_i^2 - n\bar{y}^2$$

[$\because \bar{\hat{y}} = \bar{y}$ from (a)]

Now, part (c) follows from part (b) by first subtracting $n\bar{y}^2$ from both sides and then using the above.

(d) By definition, $R^2 \geq 0$. Also, clearly from (c), $R^2 \leq 1$.
 $R^2 = 0 \Leftrightarrow \hat{y}_i = \bar{y} \quad \forall i$, i.e., the model gives a constant prediction.

$$R^2 = 1 \Leftrightarrow \sum_{i=1}^n (y_i - \hat{y}_i)^2 = 0$$

$\Leftrightarrow y_i = \hat{y}_i \quad \forall i$, i.e., the model fits the data exactly.

$$\begin{aligned} \text{(e)} \quad \hat{y}_i - \bar{y} &= \bar{y} + \hat{\beta}(x_i - \bar{x}) - \bar{y} = \hat{\beta}(x_i - \bar{x}) \\ \therefore \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 &= \hat{\beta}^2 \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{\left\{ \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right\}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ \Rightarrow \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} &= \frac{\left\{ \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \right\}^2}{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2} = r_{xy}^2. \end{aligned}$$

Ex3 We obtained $\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$.

It is easy to obtain $\hat{\gamma} = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2}$.

$$\text{Now, } \hat{\beta} = \frac{\sum_{i=1}^n x_i y_i - n \bar{x} \bar{y}}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$= \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \left(\frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2} \right) - \frac{n \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} \bar{y}$$

$$= c \hat{\gamma} + d \bar{y}$$

$$c = \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{\sum_{i=1}^n x_i^2}{\sum_{i=1}^n x_i^2 - n \bar{x}^2} \geq 1$$

(b) $\hat{\beta} = \hat{\gamma}$ if $c = 1$ and $d = 0$

This holds if $\bar{x} = 0 \Leftrightarrow \sum_{i=1}^n x_i = 0$.

If x_i 's sum to 0, then $\hat{\beta}$ & $\hat{\gamma}$ are the same.

(c) If x_i 's are centered, then the estimators are the same.

If x_i 's are also scaled, then $\hat{\beta} = \frac{s_x}{s_y} \hat{\gamma}$.

To see this, rewrite the formula for $\hat{\gamma}$ with x_i replaced by $\frac{x_i - \bar{x}}{s_x}$ & y_i replaced by $\frac{y_i - \bar{y}}{s_y}$.

Ex 4. (a) Verify that $AA^{-1} = A^{-1}A = I$, i.e.,

$$\begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1} A_{12} B^{-1} A_{21} A_{11}^{-1} & -A_{11}^{-1} A_{12} B^{-1} \\ -B^{-1} A_{21} A_{11}^{-1} & B^{-1} \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}$$

$$\& \begin{pmatrix} A_{11}^{-1} + A_{11}^{-1} A_{12} B^{-1} A_{21} A_{11}^{-1} & -A_{11}^{-1} A_{12} B^{-1} \\ -B^{-1} A_{21} A_{11}^{-1} & B^{-1} \end{pmatrix} \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix} = \begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}$$

(b) Again, verify the definition

$$(A + \underline{u} \underline{v}^T) \left(A^{-1} - \frac{A^{-1} \underline{u} \underline{v}^T A^{-1}}{1 + \underline{v}^T A^{-1} \underline{u}} \right) = I = \left(A^{-1} - \frac{A^{-1} \underline{u} \underline{v}^T A^{-1}}{1 + \underline{v}^T A^{-1} \underline{u}} \right) (A + \underline{u} \underline{v}^T)$$

(c) By definition:

$$\begin{pmatrix} \hat{\alpha}_1 \\ \hat{\beta}_1 \end{pmatrix} = (\tilde{X}^T \tilde{X})^{-1} \tilde{X}^T \underline{y} = \begin{pmatrix} n & \underline{1}^T X \\ X^T \underline{1} & X^T X \end{pmatrix}^{-1} \begin{pmatrix} \underline{1}^T \underline{y} \\ X^T \underline{y} \end{pmatrix} \\ = \begin{pmatrix} b_{11} & \underline{b}_{12}^T \\ \underline{b}_{21} & B_{22} \end{pmatrix} \begin{pmatrix} \underline{1}^T \underline{y} \\ X^T \underline{y} \end{pmatrix}$$

$$\Rightarrow \hat{\beta}_1 = \underline{1}^T \underline{y} \underline{b}_{21} + B_{22} X^T \underline{y}$$

The elements $\underline{b}_{11}$, $\underline{b}_{12}$, $\underline{b}_{21}$, B_{22} are obtained from (a)

$$b_{11} = \frac{1}{n_2} (n + \underline{1}^T X B^{-1} X^T \underline{1})$$

$$\underline{b}_{12}^T = -\frac{1}{n} \underline{1}^T X B^{-1}, \quad \underline{b}_{21} = -\frac{1}{n} B^{-1} X^T \underline{1}$$

$$B_{22} = B^{-1}, \quad B = X^T X - \frac{X^T \underline{1} \underline{1}^T X}{n}$$

$$\therefore B^{-1} = \left(X^T X - \frac{X^T \underline{1} \underline{1}^T X}{n} \right)^{-1} \left[\begin{array}{l} (A + \underline{u} \underline{v}^T)^{-1}, \text{ with } A = X^T X, \\ \underline{u} = \frac{X^T \underline{1}}{n}, \quad \underline{v} = -X^T \underline{1} \end{array} \right]$$

$$= (X^T X)^{-1} + \frac{(X^T X)^{-1} X^T \underline{1} \underline{1}^T X (X^T X)^{-1} / n}{1 - \underline{1}^T X (X^T X)^{-1} X^T \underline{1} / n}$$

$$= (X^T X)^{-1} + \frac{(X^T X)^{-1} X^T \underline{1} \underline{1}^T X (X^T X)^{-1}}{n - \underline{1}^T X (X^T X)^{-1} X^T \underline{1}}$$

$$\therefore B_{22} X^T \underline{y} = \left\{ (X^T X)^{-1} + \frac{(X^T X)^{-1} X^T \underline{1} \underline{1}^T X (X^T X)^{-1}}{n - \underline{1}^T X (X^T X)^{-1} X^T \underline{1}} \right\} X^T \underline{y}$$

$$= \hat{\beta}_2 + \frac{(X^T X)^{-1} X^T \underline{1} \underline{1}^T X}{n - \underline{1}^T X (X^T X)^{-1} X^T \underline{1}} \hat{\beta}_2$$

$$= (I + C) \hat{\beta}_2, \quad C = \frac{(X^T X)^{-1} X^T \underline{1} \underline{1}^T X}{n - \underline{1}^T X (X^T X)^{-1} X^T \underline{1}}$$

This is a scalar.

$$\text{Now, } \underline{b}_{21} = -\frac{1}{n} \underline{B}^{-1} \underline{X}^T \underline{1}$$

$$= -\frac{1}{n} \left\{ (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{1} + \frac{(\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{1} \underline{1}^T \underline{X}}{n - \underline{1}^T \underline{X} (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{1}} (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{1} \right\}$$

$$\therefore \underline{1}^T \underline{y} \underline{b}_{21} = -\frac{\underline{1}^T \underline{y}}{n} (\underline{I} + \underline{C}) (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{1}$$

$$= -\bar{y} (\underline{I} + \underline{C}) (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{1}$$

$$\therefore \hat{\underline{\beta}}_1 = \underline{1}^T \underline{y} \underline{b}_{21} + \underline{B}_{22} \underline{X}^T \underline{y}$$

$$= (\underline{I} + \underline{C}) \hat{\underline{\beta}}_2 - \bar{y} \underbrace{(\underline{I} + \underline{C}) (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{1}}_{\underline{d}}$$

$$(e) \quad \underline{C} = \underline{0} \quad \text{if} \quad (\underline{X}^T \underline{X})^{-1} \underline{X}^T \underline{1} \underline{1}^T \underline{X} = \underline{0}$$

This holds if $\underline{1}^T \underline{X} = \underline{0} \Leftrightarrow \sum_{i=1}^n x_{ij} = 0 \quad \forall j=1, \dots, p$
 i.e., the covariates are centered.

In This case, $\underline{d}$ is also $\underline{0}$.

$\underline{d}$ is also $\underline{0}$ if $\bar{y} = 0$.